1. Prerequisites

A knowledge of the specification for S1 and its prerequisites and associated formulae, together with a knowledge of differentiation and integration of polynomials, binomial coefficients in connection with the binomial distribution and the evaluation of the exponential function, is assumed and may be tested.

2. Examination

- First assessment: June 2020.
- The assessment is 1 hour and 30 minutes.
- The assessment is out of 75 marks.
- Students must answer all questions.
- Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
- The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.

3. Notation and formulae

Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.

Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

This is a list of formulae that students are expected to remember and which will not be included in formulae booklets.

For the continuous random variable X having probability density

function $\mathrm{f}(x)$,

$$\mathrm{P}(a < X \leqslant b) = \int_a^b \mathrm{f}(x)\,\mathrm{d}x$$

$$\mathrm{f}(x) = \frac{\mathrm{dF}(x)}{\mathrm{d}x}$$

S2.3 Unit content

What students need to learn:		Guidance
1. The Binomial and Poisson distributions		
1.1	The binomial and Poisson distributions.	Students will be expected to use these distributions to model a real-world situation and to comment critically on their appropriateness. Cumulative probabilities by calculation or by reference to tables. Students will be expected to use the additive property of the Poisson distribution – e.g. if the number of events per minute ~ Po(λ) then the number of events per 5 minutes ~ Po(5λ).
1.2	The mean and variance of the binomial and Poisson distributions.	No derivations will be required.
1.3	The use of the Poisson distribution as an approximation to the binomial distribution.	